# A robot that acts to know

## Day 2 — The CORTEX architecture: one belief, many owners

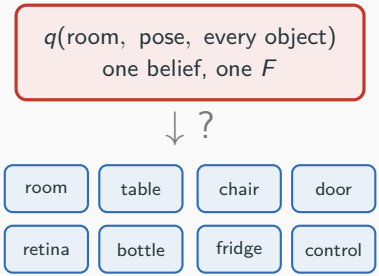
Course built on the doctoral thesis of Noé Zapata

Tuesday 1 December

Montevideo, 30 November – 4 December 2026

# Who holds the belief?

# What yesterday's equation did not say



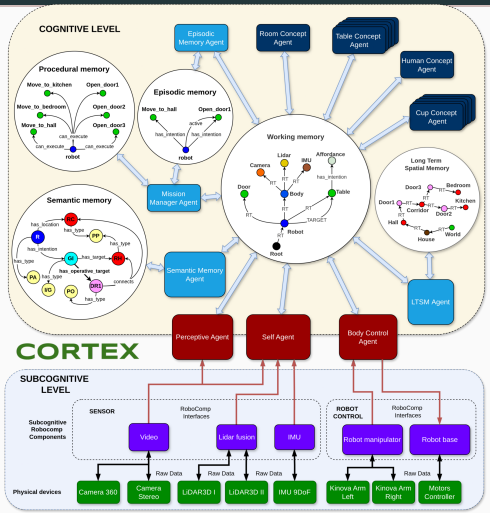

A dozen processes, sensors at 30 Hz, a robot in motion.

- **Where** is the objective computed?
- **Who** holds the beliefs?
- What happens when they must **agree** on what the world contains?

The answer is an architecture, **CORTEX**, and today's claim is that it is a factorisation of $q$.

source: cortex.tex l.107–109, l.377–378

# CORTEX



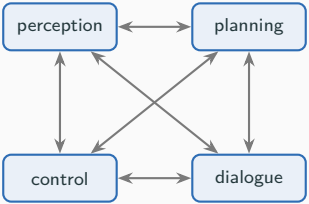

A cognitive architecture for autonomous robots: memories as graphs, and agents attached to them.

- concept agents: one class, one belief
- agents that manage the memories
- agents that reach the body
- RoboComp components

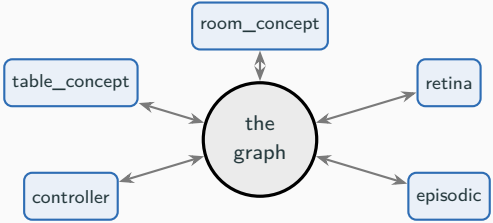
At the centre, the **working memory**: the current state, as one distributed graph. Today lives there.

source: cortex.tex l.110–113, l.154–170

# Modules that talk, or agents that share




$n(n-1)/2$ interfaces



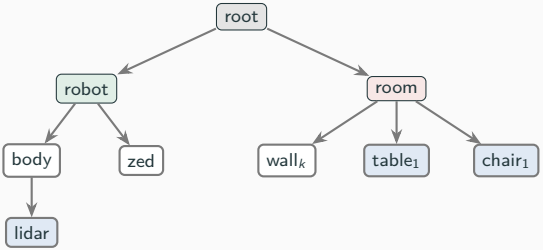

$n$ connections

There are no agent-to-agent messages. The count is the least of it: there is one place that answers "what does the robot currently believe?"

source: cortex.tex l.345–360

# The shared graph

# The working memory: geometry and meaning on the same edges




An `RT` edge: rigid transform, timestamp, **covariance**.

Nodes: a type, a name, typed attributes.

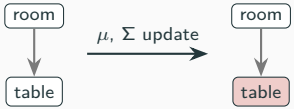
Every pose in the system is obtained by naming two frames and asking the graph to resolve the chain at a given instant:

$$^{\text{lidar}}T_{\text{table}}(t)$$

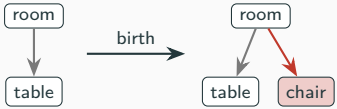
source: cortex.tex l.231–246

# Two kinds of change, two kinds of belief revision

**Non-structural**: something known has moved



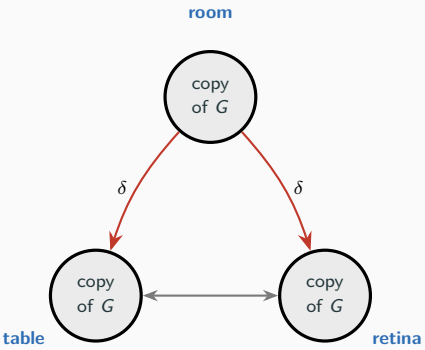

**Structural**: the world contains something new




Different agents watch for different ones. Within one structure, a belief is revised by data; when the divergence is severe and persistent, the structure itself has to change.

source: cortex.tex l.248–253, l.659–671

# Held everywhere at once

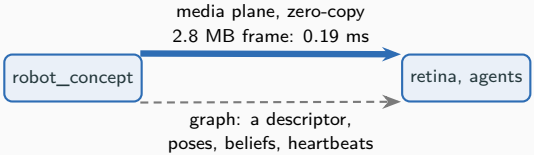



Each agent holds the whole graph and publishes **deltas**. CRDT merges need no lock.

- Merges are keyed on the writer: two agents with one id are one actor, and the result is a corrupted replica, not a conflict.
- Consistency is **eventual**: your view may briefly lag mine. Nothing may assume a global snapshot.

source: cortex.tex l.255–272

# Sensor data travels beside the graph

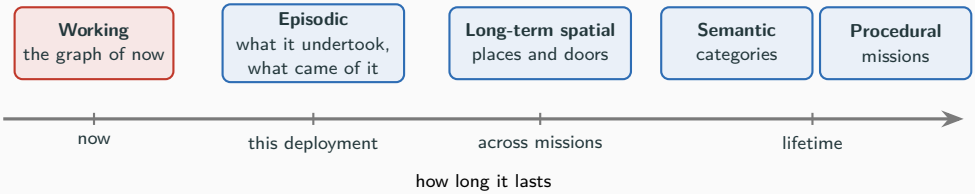



The graph is replicated to every peer. Images written into it starve the channel, and the first symptom is not slow images but a missed heartbeat: an agent declared dead because a camera was busy.

A failure in one layer that surfaces in another.

source: cortex.tex l.375–381, l.432–435

# Five memories, separated by timescale

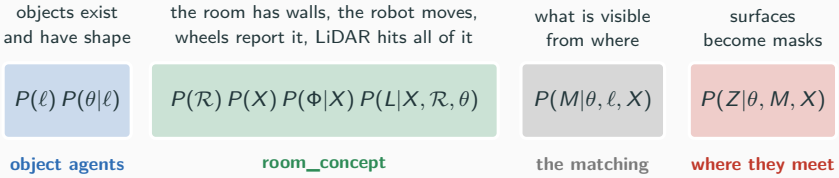



Each is a distributed graph on the same substrate, with an agent that manages it. The week lives almost entirely in the red box.

source: cortex.tex l.519–537

# One free energy, many owners

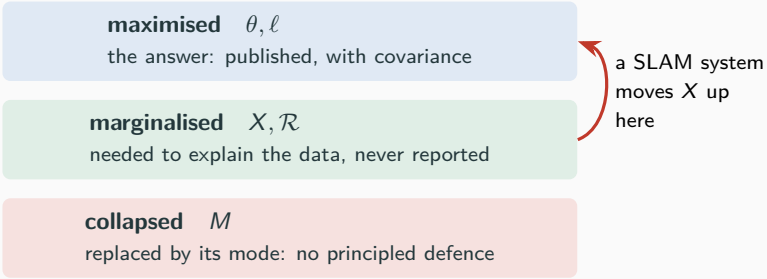
# The whole scene, written in causal order




- Objects and pose are independent *a priori*: objects do not know where the robot is.
- Once data arrive they are coupled twice: through the masks, and through the LiDAR sweep, which hits the fridge as well as the wall behind it.

source: concept_agents.tex l.255–293

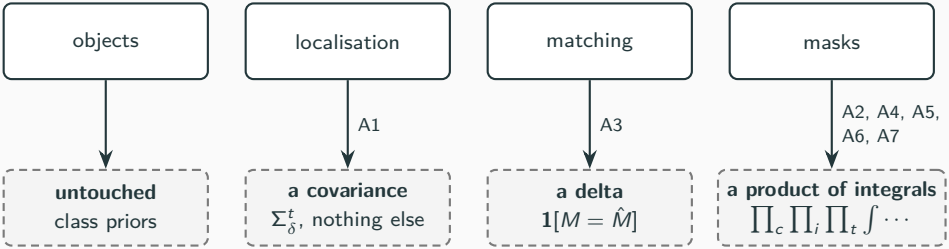
# Three ways of treating an unknown




That one reassignment separates this factorisation from a monolithic one.

source: concept_agents.tex l.395–420

# From the exact posterior to what actually runs




Seven assumptions, stated one at a time, each with its justification and its cost:

A1 objects as landmarks · A2 pose error integrates per frame · A3 hard matching · A4 classes apart · A5 instances apart · A6 geometry / existence · A7 Markov time

source: concept_agents.tex l.568–613, l.626–1019

# One process per factor

$$q(\theta, \ell, M, X, \mathcal{R}) = \underbrace{q(X, \mathcal{R})}_{\text{foreign}} \underbrace{\mathbf{1}[M = \hat{M}]}_{\text{degenerate}} \prod_{c} \prod_{i} \underbrace{q(\theta_i)}_{\mathcal{N}(\mu_i, \Sigma_i)} \underbrace{q(\ell_i)}_{\text{log-odds}}$$




Each quantity that crosses a boundary is what survived a marginalisation.

source: concept_agents.tex l.114–123, l.447–455

# What each agent minimises

$$F_i = \underbrace{\mathbb{E}_q\big[-\ln P(\bar{Z}_i \mid \theta_i, \hat{X})\big]}_{\text{accuracy: all the data enter here}} + \underbrace{D_{\mathrm{KL}}[q(\theta_i) \,\|\, P(\theta_i \mid \ell_i)]}_{\text{complexity: distance from the class prior}}$$

| complexity | accuracy | localisation | 0 |
|---|---|---|---|
| moved by the agent | | held fixed (the anchor cut) | matching inert |

Summed over instances within a class, never across classes. The localiser and the agents do not exchange in both directions.

source: concept_agents.tex l.468–516

# The smallest scene that shows the price

$n$ objects, all seen in each frame. The frame's pose error $\delta_t$ is common to all of them:

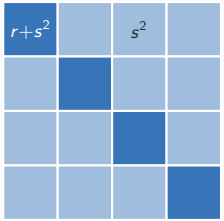
$$z_{it} = \theta_i + \delta_t + \varepsilon_{it}$$

Integrate $\delta_t$ and the frame's noise has covariance

$$C = r\, I + s^2\, \mathbf{1}\mathbf{1}^\top .$$

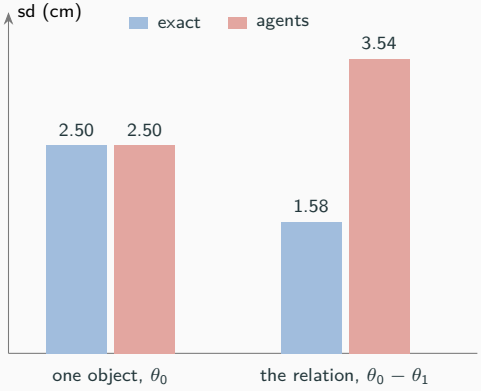
- $s^2 = 0$: the sum of the agents' $F_i$ is the global $F$, for any belief.
- $s^2 > 0$: the division is an assumption.

$C$: the $n$ sightings of one frame




source: lab 2, part C; the shared offset: concept_agents.tex l.64–70

# Each belief is right. The relation between them is wrong.

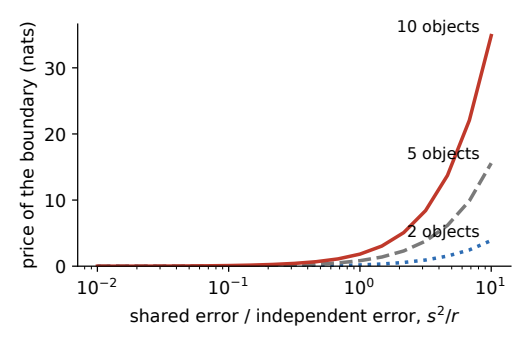



The shared error cancels in a difference. So the robot knows how far the bottle is from the table far better than it knows where either is.

Two independent beliefs on a graph cannot say that. The price is paid in relations.

source: lab 2, exercise C2 ($r = (5\,\text{cm})^2$, $s^2 = (10\,\text{cm})^2$, 20 frames, 3 objects)

# The price of a boundary, in nats




The cost of $q = \prod_i q(\theta_i)$ against the exact posterior:

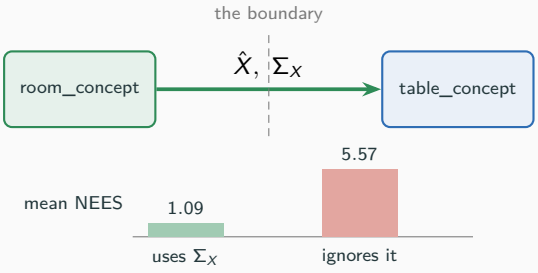
$$D_{\mathrm{KL}}[q \,\|\, p(\theta \mid z)] = F[q] - F_{\text{exact}}$$

- grows with shared error and with the number of objects;
- does not fall with more data: 2.56 nats at 10, 40 and 160 frames.

Data sharpen each factor. They cannot restore a correlation the family has no place to hold.

source: lab 2, exercise C3

# A mean is half a belief




An agent placing sightings with the published pose, and reading the edge's covariance, is honest: mean NEES $\approx 1$.

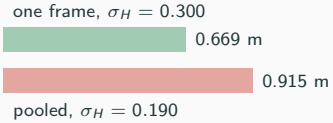
Taking the pose as truth, it is over-confident by $(r + s^2)/r = 5$.

The covariance travels with the mean, so any process can use the estimate as evidence and not as ground truth.

source: lab 2, exercise B2; concept_agents.tex l.90–95

# The architecture carries its failures on its face

**Pooling eight frames** (violates A2)




Each frame brings its own offset, the fit has nowhere to put eight of them, so they land in the shape. More confident about a worse answer.

**A cabinet labelled fridge** (cost of A4)

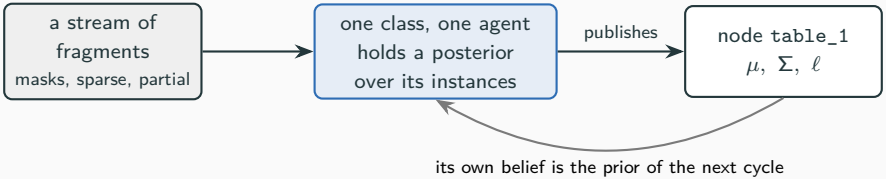



The comparison that would settle it, how well each class explains the same pixels, is never made.

source: concept_agents.tex l.73–85; agents_life_cycle.tex l.808–809

# Concept agents

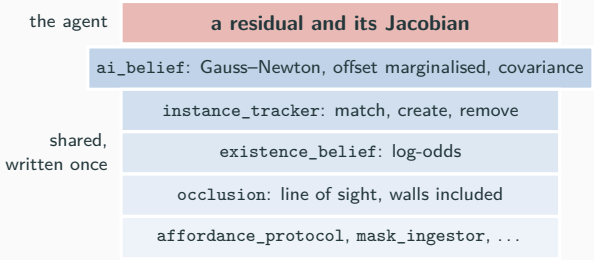
# A concept agent is neither a detector nor a service




table_concept does not detect tables. No fragment is an object: the object is a construct the agent maintains across fragments.

source: concept_agents.tex l.90–95, l.163–167

# An agent writes two things




All the Bayesian mathematics lives in the engine, once.

This is why a roster entry can be half a page: bottle needs little, table and refrigerator fill the template, and that is a fact about the objects.

source: concept_agents.tex l.534–537, l.136–138, l.1196–1203

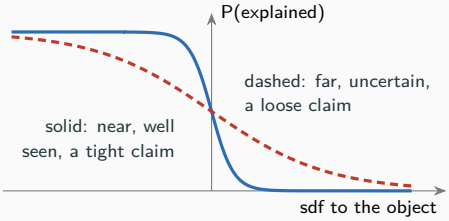
# Each publishes what only it can compute

| published on the graph | computed by | read by |
|---|---|---|
| pose $\hat{X}$ and $\Sigma_X$ | room_concept | every object agent |
| geometry $\mu_i, \Sigma_i$, in the room's frame | one object agent | residual, controller |
| existence, as log-odds | one object agent | anyone asking if it is real |
| affordances: what it offers, priced | one object agent | the controller |
| occupied and *not explained* | residual_concept | planner, other agents' births |

Conventions shared by all: geometry in the room's frame, on the RT edge; the covariance alongside; the class as an attribute; each agent recognises its own nodes by name.

source: concept_agents.tex l.1209–1216, l.1533–1544

# residual_concept: the null hypothesis



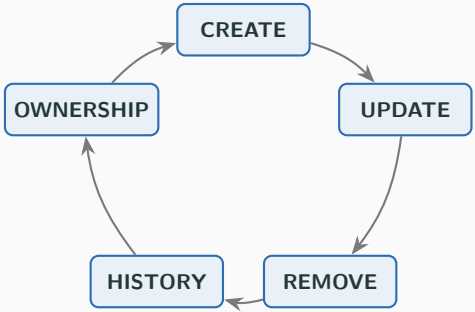

A field of what no other agent explains: the safety layer. Nothing has to *recognise* an object for the robot to avoid it.

$$P(\text{cell explained}) = \Psi\left(-\frac{\text{sdf}}{\sigma_{\text{tot}}}\right)$$

$\sigma_{\text{tot}}$ uses the object's *own published covariance*. No cutoff anywhere, and possible only because the others publish covariances.

source: concept_agents.tex l.1497–1538

# Answering to a common contract



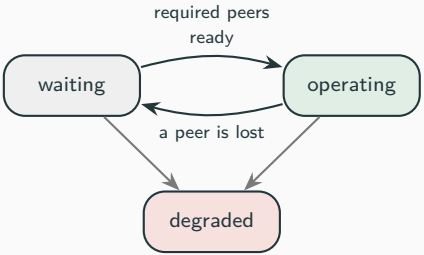

These are not an administrative layer: they are what an integrator of noisy fragments has to do in order to integrate at all.

A stage stated in a contract is a stage a new agent cannot forget.

**Tomorrow:** where beliefs come from and how they end.

source: concept_agents.tex l.169–180; CONCEPT_AGENT_LIFECYCLE.md l.7–13

# The factorisation is only true while its owners are alive




Specified in TLA$^+$ and model-checked: 8 safety and 4 liveness properties, no counterexample.

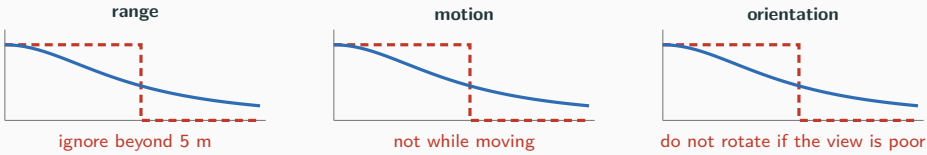
A node left behind by a dead process is a factor nothing is minimising. It looks like any other data.

A peer that is present is *running*, which is not the same as *able to serve*. Heartbeats are ordinary graph nodes.

source: cortex.tex l.732–741, l.757–775, l.780–785

# No thresholds

# A gate is what a model looks like before the modelling is done

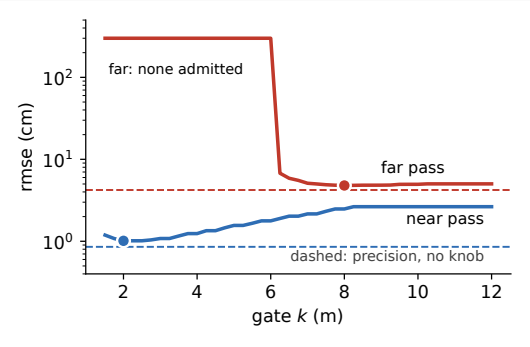



solid: the precision a frame actually has, as the covariate grows

| | |
|---|---|
| a distant view loses orientation information | common-mode covariance grows with range |
| correlated points stop shrinking $\sigma$ | the offset is marginalised in closed form |
| ego-motion down-weights a mask | a motion variance is added to the observation |

source: concept_agents.tex l.552–562

# A gate is a parameter the scene can invalidate



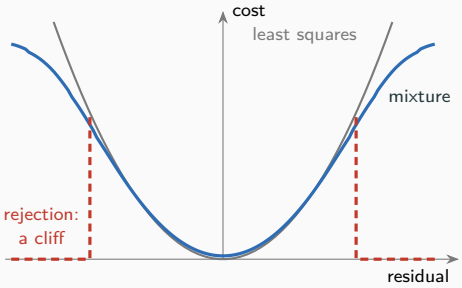

A robot passes an object, $\sigma$ growing with range.

| | near | far |
|---|---|---|
| best gate | 2.0 m | 8.0 m |
| rmse there | 1.01 cm | 4.79 cm |
| precision | 0.86 cm | 4.22 cm |

The gate tuned on the near pass admits nothing on the far one: 3 m of error. The precision has no knob.

source: lab 2, part D

# An outlier costs a bounded amount, and nobody rejects it



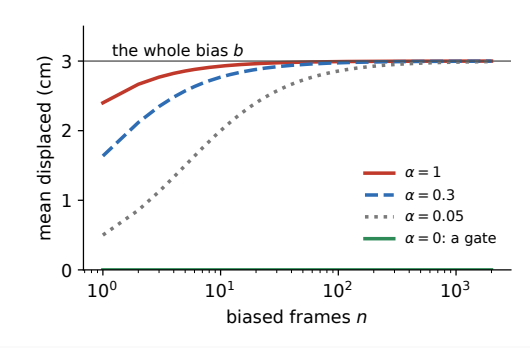

Each mask point is explained by the object's surface, by uniform clutter, or by a known wall.

$$-\ln\left(\pi_s\, p_s + \pi_c\, p_c + \pi_w\, p_w\right)$$

A point that fits nothing costs a finite amount and stops pulling. The bound is a *modelled probability*, not a rejection threshold.

source: concept_agents.tex l.518–524

# The gate that survives




A run of frames sharing one bias $b$:

$$\Delta\mu_n = \frac{n\alpha\Lambda}{\Lambda_0 + n\alpha\Lambda}\, b \xrightarrow[n\to\infty]{} b$$

Attenuating by $\alpha$ changes the rate, never the limit. Only $\alpha = 0$ bounds it.

A variance is the right answer for errors that average. This one has no coordinate in the model.

source: agents_life_cycle.tex l.783–791; lab 2, part E

# The rule is: declare it, justify it, or remove it

| threshold | verdict |
|---|---|
| admissibility | **should** be one: the continuous alternative provably cannot bound the bias |
| birth | **should not**: a streak standing in for nats; first to change |
| peer grace period | works, and is a **debt**: liveness is a belief too |

**Measured: a covariate must not be circular**

The chair's obliquity, computed from the belief's own yaw, read $\cos = 0.001$ and froze a yaw that was about $45^\circ$ wrong: a wrong estimate declared its own error unobservable.

source: agents_life_cycle.tex l.1276–1282; cortex.tex l.796–804; CONCEPT_AGENT_LIFECYCLE.md l.75–79

# Lab

# Lab 2: agents on a shared graph

`labs/day2.ipynb`, about two hours.

1. **The graph**: an ownership contract, and a sweep for the nodes of dead processes.
2. **Agents that only read and write**: an object agent that reads the robot's belief from the edge. Then remove the covariance and measure the NEES.
3. **One $F$, many owners**: when the sum of $F_i$ is the global $F$; the marginal against the relation; the price of the boundary in nats.
4. **No thresholds**: a range gate against a precision, on two passes.
5. **The gate that survives**: a bias no precision can bound.

# Tomorrow

Today the agents and the graph were given.
Tomorrow the beliefs on it have a life:

- birth as model comparison, not counting;
- absence as evidence, not a missed detection;
- forgetting as inferred volatility.

And the room and the robot's pose become beliefs like any other.

**Reading for Day 3**

Thesis chapters:

- *The Life Cycle of a Concept Agent*;
- *The Room Concept Agent and Robot Localisation*.